

Carlos Valladares

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Education

University of Santiago de Compostela – Lugo, Spain
Bachelor of Science in Robotics Engineering

Feb 2026

Work Experience

Robotics Software Engineering Intern, JdeRobot (RoboticsAcademy) – Remote

May 2026 – Present

- Develop and optimize robotics software solutions for the educational platform by deploying Docker containers and ROS 2 communication architectures
- Redesign and improve the internal node-based programming tool for visual control flow composition, optimizing modularity and abstraction in autonomous applications

Research Intern (R&D), University of Santiago de Compostela – Lugo, Spain

Feb 2025 – Jul 2025

- Processed complex biosignals using EMG sensors and motion data from IMUs to validate stabilization algorithms
- Developed predictive muscle fatigue models for a biomechanical study using Python and MATLAB

Event IT Support Contractor, Fnatic Ltd – Madrid, Spain

Apr 2025

- Coordinated the technical setup of the commercial booth and hardware infrastructure for tournament gaming stations under a fast-paced environment

Projects

Navigation2 (Nav2) Open Source Contribution

- Collaborated on the standard ROS 2 navigation stack applying agile methodologies and distributed version control
- Refactored lifecycle managers in C++ using the updated API to enhance overall system robustness

Sim-to-Real Data Generation & Visual Detection

- Trained computer vision models (YOLO) reaching high accuracy in real-time object detection using synthetic data generated within NVIDIA Isaac Sim

Design & Control of a 6-DOF Robotic Manipulator

- Directed the end-to-end mechatronic fabrication through CAD design, 3D printing, and physical assembly of structural components
- Stabilized embedded hardware by implementing real-time PID control loops and integrating UART, I2C, and SPI serial buses

3D Interactive Portfolio & Web Visualizer

github

- Developed an interactive web environment integrating React 19 and Three.js (React Three Fiber) to optimize real-time rendering and loading of 3D asset geometries
- Automated the continuous integration pipeline using GitHub Actions for compilation, artifact synchronization, and automated deployment to Vercel

Skills

Languages: Python, C, C++, Bash, MATLAB

Robotics & AI: ROS 2, Nav2, MoveIt, PyTorch, OpenCV, YOLO, ArduPilot, PX4, MAVLink, MAVSDK

Systems & Tools: Linux (Ubuntu), Docker, Git, NVIDIA Isaac Sim, GitHub Actions, Vercel, React, Three.js

Hardware: Microcontrollers, Serial Communications (UART, I2C, SPI), 3D Printing, CAD Design